

Student Performance & Academic Analytics Platform

A full-stack academic analytics system where **students track performance**, **instructors analyze class outcomes**, and **administrators monitor department-level trends**. The platform combines **interactive dashboards**, **academic reporting**, and **AI/ML-driven insights** to improve academic performance and decision-making.

This scope is still **moderate-sized**, but now includes two advanced features:

- **ML-based performance prediction**
- **Personalized recommendation engine**

These additions will make the project significantly stronger for your portfolio without turning it into an unmanageable enterprise system.

1. Authentication & Role-Based Access

Purpose

Secure system access with different dashboards based on user roles.

Roles

- Student
- Instructor
- Admin

Features

- Login/logout
- JWT authentication
- Protected API routes
- Role-based authorization
- Session persistence

Example

A student logs in and sees personal academic analytics.

An instructor logs in and sees class performance dashboards.

Why this matters

Provides real-world system structure and security.

2. Student Profile Management

Purpose

Maintain academic and personal student information.

Features

Students can view:

- Name
- Student ID
- Department
- Semester
- Batch
- Enrolled courses
- Total completed credits

Optional editing

Students can update:

- Profile picture
- Phone number
- Bio/basic details

Why this matters

Forms the foundation for analytics and reports.

3. Course Enrollment Management

Purpose

Manage student-course relationships.

Features

- View enrolled courses
- Semester-wise course list
- Instructor assignment
- Credit information
- Course history

Example

Semester 5:

- Algorithms
- Operating Systems
- Database Systems

Why this matters

Enrollment relationships power attendance, GPA, and analytics.

4. Attendance Management & Analytics

Purpose

Track attendance and identify patterns.

Instructor Features

- Mark attendance
- Bulk attendance submission
- Edit attendance

Student Features

- View attendance percentage
- Attendance per course
- Attendance history

Dashboard Analytics

- Attendance heatmaps
- Monthly attendance trends
- Course-wise attendance comparison

Example Insight

Attendance dropped by 10% in Database Systems this month.

Why this matters

Excellent feature for interactive visualizations.

5. Grade & Assessment Management

Purpose

Track academic performance across assessments.

Assessment Types

- Quiz
- Assignment
- Midterm
- Final Exam

Instructor Features

- Upload marks
- Edit grades
- Bulk grade entry

Student Features

- View scores
- Compare performance

- Subject-wise marks

Dashboard Analytics

- Grade progression
- Subject comparison
- Assessment breakdown

Example

Performance in Algorithms:

Quiz → 75

Midterm → 82

Final → 88

Why this matters

Core academic data source.

6. GPA & Academic Performance Analytics

Purpose

Automatically calculate and visualize academic performance.

Features

Automatic:

- Semester GPA
- CGPA
- Credit-weighted GPA

Dashboard Widgets

- GPA trend over semesters
- Highest/lowest GPA semester
- Best performing subjects
- Weakest subjects
- Academic progress cards

Example Insights

Your GPA improved by 0.34 compared to last semester.

Why this matters

This becomes the platform's primary analytics feature.

7. Instructor Analytics Dashboard

Purpose

Enable instructors to analyze class performance.

Features

View:

- Class average
- Grade distribution
- Attendance trends
- Top-performing students
- Underperforming students

Dashboard Visualizations

- Histograms
- Performance trends
- Attendance graphs
- Subject performance analytics

Example Insight

28% of students scored below 60% in Algorithms.

Why this matters

Makes the system feel professional and analytics-driven.

8. Semester Comparison Analytics

Purpose

Allow students to compare academic progress over time.

Features

Compare:

- GPA across semesters
- Attendance changes
- Subject performance
- Credit completion

Dashboard Visualizations

- Line charts
- Semester comparison cards
- Performance progression graphs

Example Insight

Semester 5 performance improved by 12% compared to Semester 4.

Why this matters

Strong feature for polished dashboard experiences.

9. Academic Reports Generation

Purpose

Generate downloadable academic reports.

Report Types

- Semester transcript
- Performance summary
- Attendance report

- Semester analytics report

Export Format

- PDF

Included Data

- GPA
- Grades
- Attendance
- Academic insights
- Subject performance

Example

A downloadable semester report with charts and summaries.

Why this matters

Adds a highly professional touch.

10. Admin Department Analytics

Purpose

Provide department-level academic monitoring.

Features

Admin can view:

- Department GPA averages
- Attendance statistics
- Pass/fail ratio
- Top performing courses
- Weakest courses

Dashboard Visualizations

- KPI cards

- Department comparison charts
- Trend analytics

Example Insight

Database Systems has the highest failure rate this semester.

Why this matters

Demonstrates multi-level system architecture.

11. ML-Based Performance Prediction

Purpose

Predict students who may struggle academically.

Goal

Identify students at risk **before failure happens**.

Prediction Inputs

- Attendance percentage
- Previous GPA
- Quiz marks
- Assignment performance
- Midterm scores
- Semester trends

Outputs

Examples:

Risk Level: High

Probability of poor performance: 82%

Dashboard Features

- Risk score card

- Predicted performance trend
- Student risk categories

Instructor View

Instructors can identify:

- High-risk students
- Students likely to fail
- Students needing intervention

Possible ML Models

- Logistic Regression
- Random Forest
- XGBoost

Why this matters

This becomes your strongest “advanced backend” feature.

12. Personalized Recommendation Engine

Purpose

Provide students with tailored academic suggestions.

Goal

Improve student outcomes using data-driven recommendations.

Recommendation Logic

Based on:

- Low attendance
- Poor grades
- Weak subjects
- Semester trends
- ML prediction results

Example Recommendations

Your attendance in Operating Systems is low. Aim for at least 80%.

You struggle with recursion problems in Algorithms. Practice more graph traversal questions.

Your GPA dropped this semester. Consider balancing course workload.

Recommendation Types

- Study recommendations
- Attendance improvement suggestions
- Subject-specific guidance
- Academic performance tips

Version 1 Implementation

Start with a **rule-based recommendation system**.

Example:

IF attendance < 70%
AND grade < B
THEN recommend intervention

Later, upgrade to AI-generated recommendations.

Why this matters

Makes the project feel intelligent and student-centric.

Final Feature Summary

Authentication & System

1. Authentication + role-based access

Student Features

2. Student profile management

3. Course enrollment tracking
4. Attendance analytics
5. Grade & assessment tracking
6. GPA & academic analytics
7. Semester comparison dashboard
8. Academic report generation
9. Personalized recommendations

Instructor Features

10. Attendance management
11. Marks management
12. Instructor analytics dashboard
13. Student risk prediction dashboard

Admin Features

14. Department analytics

AI/ML Features

15. ML-based performance prediction
16. Recommendation engine

Project Complexity

Frontend: 8/10

Backend: 8/10

Database Design: 8/10

This version is still very achievable, but now looks much closer to a **serious production-grade academic analytics platform** rather than a simple university management system.